Trusting the Checker

General Agents, Chapter 11/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/trusting-the-checker

EN: The whole system rests on one word: 'PASS'. This chapter turns trust in your reviewer into a measured, defended number -- the evals discipline, with small files + a shell script + jq, no framework needed.
Roman Urdu: Poora system ek lafz 'PASS' pe khara hai. Yeh chapter reviewer par trust ko measured, defended number banata hai -- evals discipline, chhoti files + shell script + jq se, koi framework zaroori nahi.

1. Test vs Eval + Golden Set

- Test = deterministic, verifies a specific property. Eval = estimates how well a probabilistic system performs on representative cases, usually across repeated runs. Metric: pass rate.
- **Golden set sourcing:** failures first (real caught = provably reachable), cover categories not volume (20-40 cases), version-control the folder.
- **3 run depths:** Depth 1 Answer (final output only), Depth 2 Actions (tools/args/order), Depth 3 Trace (full run history -- catches bad PROCESS).

2. Calibrating the Judge

- Rubric = a written spec of 'good'. Anchor every score to a real example; have the judge check FACTS, not impressions.
- **Grade the grader:** sample 20 items, blind-grade them yourself, compare to the judge's verdicts. The worst cell in the confusion matrix is FALSE PASS (bad work that shipped anyway).

3. Regression, Drift, Goodhart's Law

- The golden set IS the harness's regression suite -- rerun it before any harness change ships, compare to baseline.
- **Drift:** the judge's underlying model updates quietly -- defend with scheduled (nightly/weekly) full runs and loud alerts on any drop.
- **Goodhart's Law:** once a measure becomes the target, it stops being a good measure. Defenses: sealed hold-outs (never tuned against), refresh from real production failures, never let the agent see the answer key.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Test vs Eval	Test = deterministic property check; Eval = probabilistic performance estimate over repeated runs.	Test = deterministic property check; Eval = repeated runs par probabilistic performance estimate.
Golden set	A version-controlled set of real, categorized failure cases used to score a checker.	Version-controlled set of real, categorized failure cases, checker score karne ke liye.
False pass	The worst confusion-matrix cell -- bad work the checker approved.	Sabse bura confusion-matrix cell -- bura kaam jo checker ne approve kiya.
Grade the grader	Blind-grading a sample yourself and comparing to the judge's verdicts.	Khud blind-grade karna aur judge ke verdicts se compare karna.
Goodhart's Law	A measure optimized-against stops being a good measure.	Jis measure ko optimize kiya jaye, wo achhi measure rehna band kar deti.

Quick Self-Check

Question	Answer
What's the difference between a test and an eval?	A test deterministically verifies one property; an eval estimates how well a probabilistic system performs across representative, often repeated, runs.
Why source golden-set cases from real failures first?	A real caught failure proves that case is reachable in practice -- not a hypothetical edge case.
Which confusion-matrix cell matters most, and why?	False pass -- it means bad work was approved and shipped, the most costly kind of checker error.
What's the defense against judge drift?	Run the full golden set on a schedule (nightly/weekly), alert loudly on any score drop, and recalibrate the judge when its underlying model changes.
How do sealed hold-outs defend against Goodhart's Law?	They're never used for tuning, so a gap between tuned-set and hold-out scores reveals when the team is gaming the measure rather than improving the system.

Where This Leads / Sources

Leads to: Eval-Driven Development (Mode 2), Leaving the Laptop. Full 8 practice-project eval builds in 06-practice-projects.

Source: docs/trusting-the-checker/ (this repo), content originally via Zia Tutor AI connector.